Q1

The change in value is $N(0,\sigma^2)$ with $\sigma = 20$ mil

Var. $F(L) = \sigma \Phi^{-1}(\alpha)$, hence $VaR_u(L) = \sigma \Phi^{-1}(u)$

substituting $u = \Phi(w)$, $du = \varphi(w)\,dw$ and using $\varphi'(w) = -w\,\varphi(w)$

$$ES_\alpha(L) = \frac{\sigma}{1-\alpha}\int_\alpha^1 \Phi^{-1}(u)\,du = \frac{\sigma}{1-\alpha}\int_{\Phi^{-1}(\alpha)}^{\infty} w\,\varphi(w)\,dw =$$

$$= \frac{\sigma}{1-\alpha}\left[-\varphi(w)\right]_{\Phi^{-1}(\alpha)}^{\infty} = \sigma\,\frac{\varphi(\Phi^{-1}(\alpha))}{1-\alpha}$$

at $\alpha = 0.99$, $\Phi^{-1}(0.99) = 2.3263$,

$\varphi(2.3263) = e^{-2.3263^2/2}/\sqrt{2\pi} = 0.02665$

$VaR_{0.99} = 20 \times 2.3263 = 46.53$ mil $\quad ES_{0.99} = 20 \times \frac{0.02665}{0.01} = 53.30$ mil

2.

For each L_i has $F_{L_i}(l) = 0.98$ on $[0,10)$ and $[10,\infty)$

By independance

$P(L_1 + L_2 = 0) = 0.98^2 = 0.9604$, $P(L_1 + L_2 = 10) = 2(0.98)(0.02) = 0.0392$

$P(L_1 + L_2 = 20) = 0.02^2 = 0.0004$

a) $F_{L_i}(0) = 0.98 \geq 0.98$ and $F_{L_i}(L) = 0$ for $L < 0$, so $VaR_{0.98}(L_1) = 0$. For the sum, $F_{L_1+L_2}(L) \leq 0.9604 < 0.98$ for $L < 10$ and $F_{L_1+L_2}(10) = 0.9996 \geq 0.98$ so $VaR_{0.98}(L_1+L_2) = 10$ Therefore

$$VaR_{0.98}(L_1+L_2) = 10 > 0 = VaR_{0.98}(L_1) + VaR_{0.98}(L_2)$$

b) $VaR_{0.99}$?

Now $F_{L_i}(L) \leq 0.98 < 0.99$ for $L < 10$ and $F_{L_i}(10) = 1$, so $VaR_{0.99}(L_i) = 10$. For the sum, $F_{L_1+L_2}(10) = 0.9996 \geq 0.99$ for $L < 10$, so $VaR_{0.99}(L_1+L_2) = 10$

So:

$VaR_{0.99}(L_1+L_2) = 10 \leq 20 = VaR_{0.99}(L_1) + VaR_{0.99}(L_2)$ is therefore subadditive at this level.

c) $ES_{0.98}$?

$E[L \mid L \geq VaR]$ is not the ES

$$VaR_u(L_i) = \begin{cases} 0 & u \leq 0.98 \\ 10 & u > 0.98 \end{cases} \qquad VaR_u(L_1+L_2) = \begin{cases} 0 & u \leq 0.9604 \\ 10 & 0.9604 < u \leq 0.9996 \\ 20 & u > 0.9996 \end{cases}$$

$$ES_{0.98}(L_i) = \frac{1}{0.02}\int_{0.98}^{1} 10 \, du = 10$$

$$ES_{0.98}(L_1+L_2) = \frac{1}{0.02}(10\times 0.0196 + 20\times 0.0004) = \frac{0.204}{0.02} = 10.2$$

$$ES_{0.98}(L_1) = ES_{0.98}(L_2) = 10 \qquad ES_{0.98}(L_1+L_2) = 10.2 \leq 20$$

3.

On default the loss is $V \sim \mathcal{U}(0,10)$, otherwise is -0.2

a) VaR of loan 1.

$L_1 = -0.2$ with prob 0.9875 and $L_1 = V$ with prob 0.0125.

For $0 \leq l \leq 10$,

$$F_{L_1}(l) = 0.9875 + 0.0125\frac{l}{10}$$

and $F_{L_1} = 0.9875 < 0.99$ to the left of 0, so VaR is in this continuous increasing part:

$$0.9875 + 0.00125l = 0.99 \iff l = \frac{0.0025}{0.00125} = 2$$

$$VaR_{0.99}(L_1) = 2 \text{ mil}$$

b) $VaR_{0.99}$ of portfolio

With no default $L_1 + L_2 = -0.4$ with exactly one default $L_1 + L_2 = V - 0.2 \in (-0.2, 9.8)$. For $-0.2 \leq L \leq 9.8$

$$F_{L_1+L_2}(L) = 0.975 + 0.025\,\frac{L+0.2}{10}$$

and $F_{L_1+L_2} = 0.975 < 0.99$ to the left of -0.2, so

$$0.975 + 0.0025(L+0.2) = 0.99 \iff L+0.2 = \frac{0.015}{0.0025} = 6 \iff L = 5.8$$

$VaR_{0.99}(L_1+L_2) = 5.8$ mil since $5.8 > 2+2 = 4$ VaR is not subadditive

c) $ES_{0.99}$

Both VaRs sit in a continuous part of the distribution with $P(L \geq VaR) = 0.01$, so $ES_{0.99} = E[L \mid L \geq VaR_{0.99}]$.
Given $L_1 \geq 2$ loan 1 has defaulted and $L_1 = V$ is uniform on $[2, 10]$.

Given $L_1 + L_2 \geq 5.8$, exactly one loan has defaulted with $V \geq 6$, so the loss $V - 0.2$ is uniform on $[5.8, 9.8]$, therefore

$$ES_{0.99}(L_1) = \frac{2+10}{2} = 6 \text{ mil} \qquad ES_{0.99}(L_1+L_2) = \frac{5.8+9.8}{2} = 7.8 \text{ mil}$$

$7.8 \leq 6+6$: ES is subadditive on the same portfolio where VaR was not.

4.

Weak monotonicity:

If $L_1 \leq L_2$ a.s. then $\{L_2 \leq l\} \subseteq \{L_1 \leq l\}$ so $F_{L_2}(l) \leq F_{L_1}(l)$ for all l.

Hence $\{l : F_{L_2}(l) \geq \alpha\} \subseteq \{l : F_{L_1}(l) \geq \alpha\}$ and infimum of the smaller set is larger:

$$VaR_\alpha(L_1) \leq VaR_\alpha(L_2)$$

So the claim is:

If $L_1 < L_2$ a.s. and $\alpha < 1$ then $VaR_\alpha(L_1) < VaR_\alpha(L_2)$

Let $q_1 = VaR(L_1)$, finite because $\alpha < 1$ ($F_{L_1} \to 1 > \alpha$ at $+\infty$ and $\to 0 < \alpha$ at $-\infty$).

By definition of infimum, $F_{L_1}(l) < \alpha$ for all $l < q_1$, so $\mathbb{P}(L_1 < q_1) = \lim_{l \uparrow q_1} F_{L_1}(l) \leq \alpha$.

On $\{L_2 \leq q_1\}$ we have $L_1 < L_2 \leq q_1$,

so $\{L_2 \leq q_1\} \subseteq \{L_1 < q_1\}$ and $F_{L_2}(q_1) \leq \mathbb{P}(L_1 < q_1) \leq \alpha$.

F_{L_2} is right-continuos and non-decreasing, so $\{l : F_{L_2}(l) \geq \alpha\} = [VaR_\alpha(L_2), \infty]$. If $\mathbb{P}(L_2 < q_1) < \alpha$,

Then $F_{L_2}(q_1) < \alpha$, so $q_1 \notin [VaR(L_2), \infty)$, or

$$VaR_\alpha(L_1) = q_1 < VaR_\alpha(L_2).$$

5.

a) $L \sim Exp\left(\frac{1}{\theta}\right)$: $F(l) = 1 - e^{-l/\theta}$ for $l \geq 0$, continuous and strictly increasing

VaR: $1 - e^{-l/\theta} = \alpha \iff e^{-l/\theta} = 1 - \alpha \iff l = -\theta \ln(1-\alpha)$, so

$$VaR_\alpha(L) = \theta \ln\left(\frac{1}{1-\alpha}\right) \quad \text{and} \quad VaR_u = -\theta \ln(1-u)$$

ES:

With $v = 1 - u$ and $\int (-\ln v)\, dv = v - v \ln v$ and $v \ln v \to 0$ as $v \downarrow 0$

$$ES_\alpha(L) = \frac{1}{1-\alpha} \int_\alpha^1 (-\theta \ln(1-u))\, du = \frac{\theta}{1-\alpha} \int_0^{1-\alpha} (-\ln v)\, dv =$$

$$= \frac{\theta}{1-\alpha} \left[(1-\alpha) - (1-\alpha)\ln(1-\alpha)\right] = \theta\left(1 - \ln(1-\alpha)\right)$$

$$ES_\alpha(L) = \theta\left(1 + \ln\left(\frac{1}{1-\alpha}\right)\right) = VaR_\alpha(L) + \theta$$

b)

$L \sim \text{Pareto}(1, 1/\varepsilon)$

$F(\tau) = 1 - \tau^{-1/\varepsilon}$ for $\tau > 1$, continuous and strictly increasing with $\mathbb{P}(L > \tau) = \tau^{-1/\varepsilon}$

VaR:

$1 - \tau^{-1/\varepsilon} = \alpha \iff \tau^{-1/\varepsilon} = 1 - \alpha \iff \tau = (1-\alpha)^{-\varepsilon}$, so

$\text{VaR}_\alpha(L) = (1-\alpha)^{-\varepsilon}$ and $\text{VaR}_u(L) = (1-u)^{-\varepsilon}$

ES

with $v = 1 - u$

$$ES_\alpha(L) = \frac{1}{1-\alpha}\int_\alpha^1 (1-u)^{-\varepsilon}\,du = \frac{1}{1-\alpha}\int_0^{1-\alpha} v^{-\varepsilon}\,dv$$

for $\varepsilon < 1$: $\int_0^{1-\alpha} v^{-\varepsilon}\,dv = \frac{(1-\alpha)^{1-\varepsilon}}{1-\varepsilon}$ since $v^{1-\varepsilon} \to 0$, so

$$ES_\alpha(L) = \frac{(1-\alpha)^{-\varepsilon}}{1-\varepsilon} = \frac{\text{VaR}_\alpha(L)}{1-\varepsilon}$$

for $\varepsilon \geq 1$:

$\int_g^{1-\alpha} v^{-\varepsilon}\,dv$ equals $\ln(1-\alpha) - \ln g$ for $\varepsilon = 1$ and

and $\frac{(1-\alpha)^{1-\varepsilon} - g^{1-\varepsilon}}{1-\varepsilon}$ for $\varepsilon > 1$ both $\to +\infty$ as $g \downarrow 0$

so

$ES_\alpha(L) = +\infty$ for every $\alpha < 1$ while $VaR_\alpha(L) = (1-\alpha)^{-\varepsilon} < \infty$

Also $E[L] = \frac{1}{\varepsilon}\int_1^\infty s^{-1/\varepsilon}\, ds$ is finite only for $\varepsilon < 1$

- $VaR_\alpha = \exp\left(\varepsilon \ln\left(\frac{1}{1-\alpha}\right)\right)$ growths exponentially in ε, and as α powers of $\frac{1}{1-\alpha}$ rather than its logarithm as in (a).

- $ES_\alpha / VaR_\alpha = \frac{1}{1-\varepsilon}$ does not depend on α

- For $\varepsilon \geq 1$ the VaR is finite but ES is infinite.

Var is prob of exceeding threshold and size of exceedance has infinite mean.